(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: CIC209 ***Subject: Data Structures***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions including Q.No.1 which is compulsory. Select One question from each unit. Assume missing data, if any.

Q1 Attempt the following: **(5x3=15)**

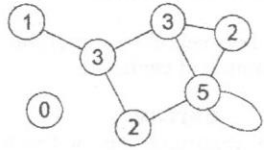
a) What do you understand by underflow and overflow in stack and queue data structures?

b) Why is it required to analyze the time complexity and space complexity of an algorithm?

c) Write short note on Bucket Hashing.

d) Write an algorithm to count the number of nodes of a binary tree.

e) Represent the following graph using Adjacency Matrix and Adjacency List representations:



UNIT-I

Q2 a) Convert the following infix expression into postfix expression using Stack: **(5)**

(A + B) / C * (D – E)^F – G

b) Write an algorithm to insert an element in the sorted doubly linked list. **(5)**

c) Write short note on double ended queue (deque). Can we implement stack and queue data structures using deque? Comment. **(5)**

Q3 a) What are the different classifications of data structures? Explain briefly all types of data structures. **(5)**

b) Consider a two-dimensional array or integers A [-11..10, 9..20]. Each element requires 4 bytes of memory storage. If the base address of the Array A is 100. You are required to derive the formula to locate the address of each element then determine the location (i.e. memory address) of A[5][15]. **(5)**

c) Explain similarities and dissimilarities between stack and queue data structures. Also list two applications of each. **(5)**

UNIT-II

Q4 a) Step by step construct the binary tree based on the below given traversals of the tree: **(5)**

Preorder: A B D C E G F H I

Inorder: B D A G E C H F I

b) What is AVL Tree? Explain different types of rotations used in AVL Tree. **(5)**

P.T.O.

CIC-209

P-1/3